

Correction Note — Step 4: Multi-Initial-Condition Evaluation (Proposed Revisions for Final Report)

1. Clarification of “Physical Stability” Terminology

Proposed revision: IC1, IC3, and IC4 remain bounded under the ias15 reference integrator over the 100-year evaluation window, while IC2 exhibits ejection under both models.

2. Interpretation of IC2 (Near-Equal Mass Case)

Proposed revision: IC2 confirms that the ejection event itself is not unique to SIMON, as ias15 also produces an ejection. Post-ejection trajectories differ due to chaotic phase error.

3. Interpretation of Zero NN Invocation (IC3 and IC4)

Proposed revision: For IC3 and IC4, no pairwise interactions crossed the NN threshold. Force evaluation reduces to analytic Newtonian gravity; differences from ias15 arise from the leapfrog integrator.

4. Speedup Interpretation

Proposed revision: Across bounded configurations, SIMON achieves 1.10×–1.18× speedup. The 1.56× case (IC2) is reported separately as a dynamically unstable configuration.

5. Formatting Corrections

Replace $M_{\bullet}$ → M_{sun} , E_{bullet} → E_{total} , and standardize λ notation.

6. Final Revised Summary Statement

SIMON was evaluated across four initial conditions. In the three bounded cases, SIMON maintains bounded trajectories with $\lambda = 0.128 \pm 0.028 \text{ yr}^{-1}$ and RMS errors between 1.54–3.45 AU. The near-equal-mass case produces ejection under both models, confirming physical instability.